

Machine Learning: A Comprehensive Textbook

Solutions to Chapter 2 Exercises Mathematical Foundations for Machine Learning

Solutions Manual

Exercise 2.1 — Transpose Reversal

Prove $(AB)^\top = B^\top A^\top$ using the definition $(C)_{ij} = \sum_k A_{ik} B_{kj}$.

Solution. Let $C = AB$, so $C_{ij} = \sum_k A_{ik} B_{kj}$. By definition of transpose, $(C^\top)_{ij} = C_{ji} = \sum_k A_{jk} B_{ki}$.

Now compute the (i, j) entry of $B^\top A^\top$:

$$(B^\top A^\top)_{ij} = \sum_k (B^\top)_{ik} (A^\top)_{kj} = \sum_k B_{ki} A_{jk} = \sum_k A_{jk} B_{ki}.$$

This is identical to $(C^\top)_{ij}$ term-by-term. Since i, j were arbitrary, $(AB)^\top = C^\top = B^\top A^\top$. ■

Exercise 2.2 — Projection Matrix

Show that $H = X(X^\top X)^{-1}X^\top$ is (a) symmetric, (b) idempotent, (c) has only eigenvalues 0 and 1. What is $\text{rank}(H)$?

Solution. (a) **Symmetric.** Using $(ABC)^\top = C^\top B^\top A^\top$ and $((X^\top X)^{-1})^\top = ((X^\top X)^\top)^{-1} = (X^\top X)^{-1}$ (since $X^\top X$ is symmetric, so is its inverse):

$$H^\top = (X(X^\top X)^{-1}X^\top)^\top = X((X^\top X)^{-1})^\top X^\top = X(X^\top X)^{-1}X^\top = H.$$

(b) **Idempotent.**

$$H^2 = X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top = X(X^\top X)^{-1} \underbrace{[(X^\top X)(X^\top X)^{-1}]}_{=I} X^\top = X(X^\top X)^{-1}X^\top = H.$$

(c) **Eigenvalues** $\in \{0, 1\}$. Let $Hv = \lambda v$ for eigenvector $v \neq 0$. Applying H again: $H^2v = \lambda Hv = \lambda^2 v$. But $H^2 = H$, so $Hv = \lambda^2 v$, giving $\lambda v = \lambda^2 v$, i.e. $\lambda(\lambda - 1)v = 0$. Since $v \neq 0$, $\lambda(\lambda - 1) = 0$, so $\lambda \in \{0, 1\}$.

Rank. For a symmetric idempotent matrix, $\text{rank}(H) = \text{tr}(H)$ (since H is diagonalizable with eigenvalues only 0/1, the trace counts the number of 1-eigenvalues, which equals the rank). Also

$$\text{tr}(H) = \text{tr}(X(X^\top X)^{-1}X^\top) = \text{tr}((X^\top X)^{-1}X^\top X) = \text{tr}(I_d) = d,$$

using cyclic invariance of trace, where d is the number of columns of X (assuming X has full column rank so $X^\top X$ is invertible). So $\text{rank}(H) = d$, the number of predictors — H (the “hat matrix”) projects onto the d -dimensional column space of X .

Exercise 2.3 — SVD Properties

If $A = U\Sigma V^\top$: (a) show $A^\dagger = V\Sigma^\dagger U^\top$; (b) show $\|A\|_F^2 = \sum_i \sigma_i^2$; (c) relate singular values to eigenvalues when A is square symmetric.

Solution. (a) **Pseudoinverse.** The Moore–Penrose pseudoinverse $A^\dagger$ is defined by the four Penrose conditions: $AA^\dagger A = A$, $A^\dagger AA^\dagger = A^\dagger$, $(AA^\dagger)^\top = AA^\dagger$, $(A^\dagger A)^\top = A^\dagger A$. Define $\Sigma^\dagger$ by inverting the nonzero singular values on the diagonal (and transposing shape) and set $B = V\Sigma^\dagger U^\top$. Check the conditions using $U^\top U = I$, $V^\top V = I$, and $\Sigma\Sigma^\dagger\Sigma = \Sigma$ (true entrywise since $\sigma_i\sigma_i^\dagger\sigma_i = \sigma_i$ whether $\sigma_i = 0$ or not):

$$ABA = (U\Sigma V^\top)(V\Sigma^\dagger U^\top)(U\Sigma V^\top) = U(\Sigma\Sigma^\dagger\Sigma)V^\top = U\Sigma V^\top = A.$$

$$BAB = (V\Sigma^\dagger U^\top)(U\Sigma V^\top)(V\Sigma^\dagger U^\top) = V(\Sigma^\dagger\Sigma\Sigma^\dagger)U^\top = V\Sigma^\dagger U^\top = B.$$

$AB = U\Sigma\Sigma^\dagger U^\top$: since $\Sigma\Sigma^\dagger$ is diagonal (hence symmetric), AB is symmetric ($(AB)^\top = U(\Sigma\Sigma^\dagger)^\top U^\top = AB$). Similarly $BA = V\Sigma^\dagger\Sigma V^\top$ is symmetric. All four conditions hold, so by uniqueness of the Moore–Penrose pseudoinverse, $A^\dagger = B = V\Sigma^\dagger U^\top$.

(b) **Frobenius norm.** $\|A\|_F^2 = \text{tr}(A^\top A)$. Substituting the SVD:

$$\text{tr}(A^\top A) = \text{tr}((U\Sigma V^\top)^\top (U\Sigma V^\top)) = \text{tr}(V\Sigma^\top U^\top U\Sigma V^\top) = \text{tr}(V\Sigma^\top \Sigma V^\top).$$

By cyclic invariance, $\text{tr}(V\Sigma^\top \Sigma V^\top) = \text{tr}(\Sigma^\top \Sigma V^\top V) = \text{tr}(\Sigma^\top \Sigma)$ (using $V^\top V = I$). Since $\Sigma^\top \Sigma$ is diagonal with entries σ_i^2 ,

$$\|A\|_F^2 = \text{tr}(\Sigma^\top \Sigma) = \sum_i \sigma_i^2.$$

(c) **Symmetric square A .** If $A = A^\top$ is real symmetric, the spectral theorem gives $A = Q\Lambda Q^\top$ with Q orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ real eigenvalues. Comparing to the SVD $A = U\Sigma V^\top$ with $\Sigma \geq 0$: if all $\lambda_i \geq 0$, then $U = V = Q$ and $\sigma_i = \lambda_i$ directly. In general, singular values are always non-negative, so $\sigma_i = |\lambda_i|$, and the sign is absorbed into U vs. V differing by a sign flip on columns where $\lambda_i < 0$ (i.e. $u_i = \text{sign}(\lambda_i) q_i$, $v_i = q_i$). So for symmetric A : *the singular values are the absolute values of the eigenvalues*, $\sigma_i = |\lambda_i|$.

Exercise 2.4 — Gradient Derivation

Derive $\nabla_\theta \|y - X\theta\|_2^2$ from scratch (expand the quadratic form, differentiate term by term). Set to zero to derive the Normal Equations.

Solution. Expand:

$$\|y - X\theta\|_2^2 = (y - X\theta)^\top (y - X\theta) = y^\top y - y^\top X\theta - \theta^\top X^\top y + \theta^\top X^\top X\theta.$$

Since $y^\top X\theta$ is a scalar, it equals its own transpose: $y^\top X\theta = \theta^\top X^\top y$. So

$$\|y - X\theta\|_2^2 = y^\top y - 2\theta^\top X^\top y + \theta^\top X^\top X\theta.$$

Differentiate term by term using the matrix-calculus identities $\nabla_\theta(a^\top \theta) = a$ and $\nabla_\theta(\theta^\top A\theta) = 2A\theta$ for symmetric A (here $A = X^\top X$, which is symmetric):

- $\nabla_\theta(y^\top y) = 0$ (constant in θ).
- $\nabla_\theta(-2\theta^\top X^\top y) = -2X^\top y$ (matching the form $a^\top \theta$ with $a = X^\top y$, note $\theta^\top X^\top y = (X^\top y)^\top \theta$).

- $\nabla_{\theta}(\theta^{\top} X^{\top} X \theta) = 2X^{\top} X \theta$.

Summing:

$$\nabla_{\theta} \|y - X\theta\|_2^2 = -2X^{\top} y + 2X^{\top} X \theta.$$

Setting this to zero:

$$2X^{\top} X \theta - 2X^{\top} y = 0 \iff X^{\top} X \theta = X^{\top} y \quad (\text{the Normal Equations}).$$

If $X^{\top} X$ is invertible (i.e. X has full column rank), $\theta^* = (X^{\top} X)^{-1} X^{\top} y$. The Hessian $\nabla_{\theta}^2 \|y - X\theta\|_2^2 = 2X^{\top} X \succeq 0$ confirms this is a global minimum (convex quadratic).

Exercise 2.5 — Chain Rule for a Two-Layer Network

Let $h = \sigma(W_1 x)$, $\hat{y} = w_2^{\top} h$, $L = \frac{1}{2}(\hat{y} - y)^2$. (a) $\partial L / \partial w_2$. (b) $\partial L / \partial W_1$, step by step. (c) Dimensionality of $\partial L / \partial W_1$.

Solution. Let $z = W_1 x \in \mathbb{R}^m$ so $h = \sigma(z) \in \mathbb{R}^m$, and $\hat{y} = w_2^{\top} h \in \mathbb{R}$ with $w_2 \in \mathbb{R}^m$, $x \in \mathbb{R}^d$, $W_1 \in \mathbb{R}^{m \times d}$.

(a) By the chain rule,

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_2} = (\hat{y} - y) \cdot h.$$

(Here $\partial \hat{y} / \partial w_2 = h$ since $\hat{y} = w_2^{\top} h$ is linear in w_2 .) So $\partial L / \partial w_2 = (\hat{y} - y) h \in \mathbb{R}^m$.

(b) Chain through $\hat{y} \rightarrow h \rightarrow z \rightarrow W_1$:

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial W_1}.$$

Term by term:

- $\frac{\partial L}{\partial \hat{y}} = (\hat{y} - y)$ (scalar).
- $\frac{\partial \hat{y}}{\partial h} = w_2 \in \mathbb{R}^m$ (since $\hat{y} = w_2^{\top} h$).
- $\frac{\partial h}{\partial z} = \text{diag}(\sigma'(z)) \in \mathbb{R}^{m \times m}$, since $h_k = \sigma(z_k)$ depends only on z_k .
- $\frac{\partial z}{\partial W_1}$: since $z = W_1 x$, $z_k = \sum_j (W_1)_{kj} x_j$, so $\partial z_k / \partial (W_1)_{kj} = x_j$ and $\partial z_k / \partial (W_1)_{k'j} = 0$ for $k' \neq k$.

Combine the first three factors into a vector $\delta \in \mathbb{R}^m$ (the “error signal” at the hidden layer):

$$\delta = (\hat{y} - y) w_2 \odot \sigma'(z) \in \mathbb{R}^m, \quad \delta_k = (\hat{y} - y) (w_2)_k \sigma'(z_k).$$

Then by the last factor,

$$\frac{\partial L}{\partial (W_1)_{kj}} = \delta_k \cdot x_j,$$

i.e. in matrix form,

$$\boxed{\frac{\partial L}{\partial W_1} = \delta x^{\top} = [(\hat{y} - y) w_2 \odot \sigma'(z)] x^{\top}.$$

(c) $\delta \in \mathbb{R}^m$, $x \in \mathbb{R}^d$, so $\delta x^{\top} \in \mathbb{R}^{m \times d}$ — the same shape as W_1 itself, as required for a valid gradient.

Exercise 2.6 — MLE Derivations

Derive the MLE for (a) Bernoulli(p), (b) Exponential(λ), (c) $\mathcal{N}(\mu, \sigma^2)$ (show $\hat{\sigma}^2$ is biased).

Solution. (a) Bernoulli. Likelihood for n flips with k heads: $L(p) = p^k(1-p)^{n-k}$. Log-likelihood: $\ell(p) = k \ln p + (n-k) \ln(1-p)$. Differentiate and set to zero:

$$\ell'(p) = \frac{k}{p} - \frac{n-k}{1-p} = 0 \implies k(1-p) = (n-k)p \implies k = np \implies \hat{p} = \frac{k}{n}.$$

(Second derivative $-k/p^2 - (n-k)/(1-p)^2 < 0$ confirms a maximum.)

(b) Exponential. Likelihood: $L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x^{(i)}} = \lambda^n e^{-\lambda \sum_i x^{(i)}}$. Log-likelihood: $\ell(\lambda) = n \ln \lambda - \lambda \sum_i x^{(i)}$. Setting the derivative to zero:

$$\ell'(\lambda) = \frac{n}{\lambda} - \sum_i x^{(i)} = 0 \implies \hat{\lambda} = \frac{n}{\sum_i x^{(i)}} = \frac{1}{\bar{x}}.$$

(c) Gaussian. Log-likelihood for $\mathcal{N}(\mu, \sigma^2)$:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (x^{(i)} - \mu)^2.$$

For μ : $\partial \ell / \partial \mu = \frac{1}{\sigma^2} \sum_i (x^{(i)} - \mu) = 0 \implies \hat{\mu} = \frac{1}{n} \sum_i x^{(i)} = \bar{x}$.

For σ^2 (treating σ^2 as a single variable v): $\ell = -\frac{n}{2} \ln(2\pi v) - \frac{1}{2v} \sum_i (x^{(i)} - \mu)^2$.

$$\frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{1}{2v^2} \sum_i (x^{(i)} - \mu)^2 = 0 \implies \hat{v} = \frac{1}{n} \sum_i (x^{(i)} - \hat{\mu})^2 = \hat{\sigma}^2.$$

So $\hat{\sigma}^2 = \frac{1}{n} \sum_i (x^{(i)} - \bar{x})^2$.

Bias of $\hat{\sigma}^2$. Write $x^{(i)} - \bar{x} = (x^{(i)} - \mu) - (\bar{x} - \mu)$. Then

$$\sum_i (x^{(i)} - \bar{x})^2 = \sum_i (x^{(i)} - \mu)^2 - n(\bar{x} - \mu)^2$$

(standard identity, from expanding the square and using $\sum_i (x^{(i)} - \mu) = n(\bar{x} - \mu)$). Taking expectations, using $\mathbb{E}[(x^{(i)} - \mu)^2] = \sigma^2$ and $\mathbb{E}[(\bar{x} - \mu)^2] = \text{Var}(\bar{x}) = \sigma^2/n$:

$$\mathbb{E} \left[\sum_i (x^{(i)} - \bar{x})^2 \right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2.$$

So

$$\mathbb{E}[\hat{\sigma}^2] = \mathbb{E} \left[\frac{1}{n} \sum_i (x^{(i)} - \bar{x})^2 \right] = \frac{n-1}{n} \sigma^2 \neq \sigma^2,$$

confirming $\hat{\sigma}_{\text{MLE}}^2$ is **biased downward** by a factor of $(n-1)/n$; the unbiased estimator divides by $n-1$ instead of n (Bessel's correction).

Exercise 2.7 — Regularisation as MAP

Show that MAP estimation with Gaussian prior $p(\theta) = \mathcal{N}(0, \tau^2 I)$ for logistic regression is equivalent to ℓ_2 -regularised logistic regression with $\lambda = 1/\tau^2$.

Solution. The MAP estimate maximizes the posterior $p(\theta | \mathcal{D}) \propto p(\mathcal{D} | \theta) p(\theta)$, i.e.

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \left[\log p(\mathcal{D} | \theta) + \log p(\theta) \right].$$

The Bernoulli log-likelihood for $P(y = 1 | x; \theta) = \sigma(\theta^\top x)$ is

$$\log p(\mathcal{D} | \theta) = \sum_{i=1}^n \left[y^{(i)} \log \sigma(\theta^\top x^{(i)}) + (1 - y^{(i)}) \log(1 - \sigma(\theta^\top x^{(i)})) \right].$$

The Gaussian prior log-density is

$$\log p(\theta) = \log \left[(2\pi\tau^2)^{-d/2} \exp\left(-\frac{\|\theta\|_2^2}{2\tau^2}\right) \right] = -\frac{d}{2} \log(2\pi\tau^2) - \frac{\|\theta\|_2^2}{2\tau^2}.$$

The additive constant $-\frac{d}{2} \log(2\pi\tau^2)$ does not depend on θ , so it does not affect the argmax. Hence

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \left[\log p(\mathcal{D} | \theta) - \frac{1}{2\tau^2} \|\theta\|_2^2 \right] = \arg \min_{\theta} \left[-\log p(\mathcal{D} | \theta) + \frac{1}{2\tau^2} \|\theta\|_2^2 \right].$$

Writing $J(\theta) = -\log p(\mathcal{D} | \theta)$ (the standard logistic cross-entropy loss), this is

$$\hat{\theta}_{\text{MAP}} = \arg \min_{\theta} \left[J(\theta) + \frac{\lambda}{2} \|\theta\|_2^2 \right], \quad \lambda := \frac{1}{\tau^2},$$

which is exactly ℓ_2 -regularised (ridge-penalised) logistic regression with regularisation strength $\lambda = 1/\tau^2$. A tighter prior (smaller τ^2 , i.e. stronger belief that θ is near 0) corresponds to stronger regularisation (larger λ), matching intuition.

Exercise 2.8 — KL Divergence

(a) Compute $D_{\text{KL}}(p||q)$ and $D_{\text{KL}}(q||p)$ for $p = \mathcal{N}(0, 1)$, $q = \mathcal{N}(\mu, \sigma^2)$. (b) Show $D_{\text{KL}}(p||q) \neq D_{\text{KL}}(q||p)$ for $\mu \neq 0$. (c) Explain forward vs. reverse KL geometrically for bimodal p .

Solution. (a) The general closed form for $D_{\text{KL}}(\mathcal{N}(\mu_1, \sigma_1^2) || \mathcal{N}(\mu_2, \sigma_2^2))$ is

$$D_{\text{KL}}(\mathcal{N}(\mu_1, \sigma_1^2) || \mathcal{N}(\mu_2, \sigma_2^2)) = \ln \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}.$$

(Derivation: $D_{\text{KL}}(p||q) = \mathbb{E}_p[\ln p(x) - \ln q(x)]$; expand both Gaussian log-densities and use $\mathbb{E}_p[(x - \mu_1)^2] = \sigma_1^2$, $\mathbb{E}_p[(x - \mu_2)^2] = \sigma_1^2 + (\mu_1 - \mu_2)^2$.)

With $p = \mathcal{N}(0, 1)$ ($\mu_1 = 0, \sigma_1 = 1$), $q = \mathcal{N}(\mu, \sigma^2)$ ($\mu_2 = \mu, \sigma_2 = \sigma$):

$$D_{\text{KL}}(p||q) = \ln \sigma + \frac{1 + \mu^2}{2\sigma^2} - \frac{1}{2}.$$

Swapping roles ($\mu_1 = \mu, \sigma_1 = \sigma$; $\mu_2 = 0, \sigma_2 = 1$):

$$D_{\text{KL}}(q||p) = \ln \frac{1}{\sigma} + \frac{\sigma^2 + \mu^2}{2} - \frac{1}{2} = -\ln \sigma + \frac{\sigma^2 + \mu^2}{2} - \frac{1}{2}.$$

(b) These are manifestly different functions of (μ, σ) . E.g. at $\sigma = 1$: $D_{\text{KL}}(p||q) = \frac{\mu^2}{2}$ and $D_{\text{KL}}(q||p) = \frac{\mu^2}{2}$ — equal when $\sigma = 1$! But take $\sigma = 2, \mu = 1$: $D_{\text{KL}}(p||q) = \ln 2 + \frac{1+1}{8} - \frac{1}{2} = \ln 2 - \frac{1}{4} \approx 0.443$, while $D_{\text{KL}}(q||p) = -\ln 2 + \frac{4+1}{2} - \frac{1}{2} = -\ln 2 + 2 \approx 1.307$. These differ ($0.443 \neq 1.307$), confirming asymmetry: $D_{\text{KL}}(p||q) \neq D_{\text{KL}}(q||p)$ in general (equality is a special coincidence, not the rule — e.g. it fails whenever $\sigma \neq 1$).

(c) **Geometric intuition.** Let p be bimodal (e.g. a mixture of two well-separated Gaussians) and q a single unimodal Gaussian we are fitting.

- **Forward KL**, $D_{\text{KL}}(p||q) = \mathbb{E}_p[\log(p/q)]$: the expectation is taken *under* p . Wherever $p(x) > 0$ but $q(x) \approx 0$, the ratio $p/q \rightarrow \infty$ and the integrand blows up — forward KL imposes an enormous penalty for q assigning near-zero mass to any region where p has mass. The minimizer is therefore forced to *cover* both modes of p , typically by spreading q 's single mode wide enough to overlap both bumps (or even placing q 's mean between them) — this is the classical “mean-seeking”/mass-covering behavior, at the cost of placing probability mass in the low-density valley between p 's two modes where p itself has little mass.
- **Reverse KL**, $D_{\text{KL}}(q||p) = \mathbb{E}_q[\log(q/p)]$: the expectation is under q . Now, wherever $q(x) > 0$ but $p(x) \approx 0$, the ratio blows up — reverse KL heavily penalizes q placing mass where p has none. The minimizer instead concentrates q entirely within a single one of p 's modes (whichever gives lower cost/is closer to initialization), ignoring the other mode completely — this is the classical “mode-seeking” behavior: q never straddles the valley, because that would mean assigning positive q -density to a region where $p \approx 0$.

This distinction is central to variational inference (which typically minimizes reverse KL, hence VI posteriors tend to under-cover/underestimate the true posterior's spread and can collapse onto a single mode) versus e.g. maximum-likelihood fitting (which is equivalent to minimizing forward KL between the empirical data distribution and the model).

Exercise 2.9 — Jensen's Inequality

Prove Jensen's inequality for convex ϕ and discrete X taking values $x_1, \dots, x_n$ with probabilities $p_1, \dots, p_n$. Derive KL non-negativity from Jensen.

Solution. Jensen's inequality. We prove $\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$ for convex ϕ , by induction on n (number of support points).

Base case $n = 2$: convexity of ϕ means precisely $\phi(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda\phi(x_1) + (1 - \lambda)\phi(x_2)$ for $\lambda \in [0, 1]$. Taking $\lambda = p_1$ (so $1 - \lambda = p_2$) gives $\phi(p_1 x_1 + p_2 x_2) \leq p_1\phi(x_1) + p_2\phi(x_2)$, i.e. exactly $\phi(\mathbb{E}X) \leq \mathbb{E}[\phi(X)]$.

Inductive step: suppose the inequality holds for any $n - 1$ point distribution. For n points with weights $p_1, \dots, p_n$ summing to 1, let $q = \sum_{i=2}^n p_i = 1 - p_1$ (assume $q > 0$; if $q = 0$ the claim is trivial) and define $\tilde{p}_i = p_i/q$ for $i = 2, \dots, n$, a valid probability vector over $x_2, \dots, x_n$. Write $\bar{x} = \sum_{i=2}^n \tilde{p}_i x_i = \frac{1}{q} \sum_{i \geq 2} p_i x_i$, so that $\mathbb{E}[X] = p_1 x_1 + q\bar{x}$. Then:

$$\phi(\mathbb{E}[X]) = \phi(p_1 x_1 + q\bar{x}) \leq p_1\phi(x_1) + q\phi(\bar{x}) \quad (\text{convexity, base case with two “points” } x_1, \bar{x})$$

$$\leq p_1\phi(x_1) + q \sum_{i=2}^n \tilde{p}_i\phi(x_i) \quad (\text{inductive hypothesis on } n - 1 \text{ points})$$

$$= p_1\phi(x_1) + \sum_{i=2}^n p_i\phi(x_i) = \sum_{i=1}^n p_i\phi(x_i) = \mathbb{E}[\phi(X)].$$

This completes the induction, proving $\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$ for any finite discrete X and convex ϕ . ■

KL non-negativity from Jensen. For distributions p, q over the same finite support,

$$D_{\text{KL}}(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = - \sum_x p(x) \log \frac{q(x)}{p(x)} = -\mathbb{E}_{X \sim p} \left[\log \frac{q(X)}{p(X)} \right].$$

$-\log$ is convex, so by Jensen's inequality applied to the random variable $Z = q(X)/p(X)$ (with $X \sim p$):

$$-\mathbb{E}_p \left[\log \frac{q(X)}{p(X)} \right] \geq -\log \left(\mathbb{E}_p \left[\frac{q(X)}{p(X)} \right] \right).$$

Compute the inner expectation:

$$\mathbb{E}_p \left[\frac{q(X)}{p(X)} \right] = \sum_{x: p(x) > 0} p(x) \cdot \frac{q(x)}{p(x)} = \sum_{x: p(x) > 0} q(x) \leq \sum_x q(x) = 1.$$

So $-\log(\mathbb{E}_p[q/p]) \geq -\log(1) = 0$. Combining,

$$D_{\text{KL}}(p\|q) \geq -\log(1) = 0,$$

with equality iff $\phi = -\log$ is applied at a degenerate point (Jensen is tight iff $Z = q(X)/p(X)$ is a.s. constant under p , i.e. $q(x)/p(x)$ is the same for all x in the support of p) combined with $\sum_{p(x) > 0} q(x) = 1$, which together force $p = q$ everywhere. Hence $D_{\text{KL}}(p\|q) \geq 0$ with equality iff $p = q$. ■

Exercise 2.10 — Entropy and KL

(a) Prove $H(p, q) \geq H(p)$ with equality iff $p = q$. (b) Compute cross-entropy, KL, entropy for $\hat{p} = [0.7, 0.2, 0.1]$, true $p = [1, 0, 0]$. (c) Why is the ML “cross-entropy loss” really KL up to a constant?

Solution. (a) By definition, $H(p, q) = -\sum_x p(x) \log q(x)$ and $H(p) = -\sum_x p(x) \log p(x)$, so

$$H(p, q) - H(p) = -\sum_x p(x) \log q(x) + \sum_x p(x) \log p(x) = \sum_x p(x) \log \frac{p(x)}{q(x)} = D_{\text{KL}}(p\|q) \geq 0,$$

using the KL non-negativity proven in Exercise 2.9, with equality iff $p = q$. Hence $H(p, q) \geq H(p)$, equality iff $p = q$. ■

(b) True label is one-hot, $p = [1, 0, 0]$ (class 1), prediction $\hat{p} = [0.7, 0.2, 0.1]$.

Entropy of p : $H(p) = -\sum_i p_i \log p_i = -1 \cdot \log 1 - 0 - 0 = 0$ (a degenerate/one-hot distribution has zero entropy).

Entropy of $\hat{p}$: $H(\hat{p}) = -(0.7 \log 0.7 + 0.2 \log 0.2 + 0.1 \log 0.1)$. Using natural log: $0.7 \log 0.7 \approx 0.7(-0.3567) = -0.2497$, $0.2 \log 0.2 \approx 0.2(-1.6094) = -0.3219$, $0.1 \log 0.1 \approx 0.1(-2.3026) = -0.2303$. Sum ≈ -0.8018 , so $H(\hat{p}) \approx 0.8018$ nats.

Cross-entropy $H(p, \hat{p})$: $H(p, \hat{p}) = -\sum_i p_i \log \hat{p}_i = -1 \cdot \log(0.7) - 0 - 0 = -\log(0.7) \approx 0.3567$ nats. (Only the true class's predicted probability matters, since p is one-hot.)

KL divergence: $D_{\text{KL}}(p\|\hat{p}) = H(p, \hat{p}) - H(p) = 0.3567 - 0 = 0.3567$ nats (identical to the cross-entropy here, precisely because $H(p) = 0$).

(c) In supervised classification, the “label” distribution p is (almost always) the one-hot ground-truth distribution, whose entropy $H(p) = 0$ is a fixed constant with respect to the model parameters θ (it does not depend on the model’s predictions $\hat{p}_\theta$ at all). Since $H(p, \hat{p}_\theta) = H(p) + D_{\text{KL}}(p \| \hat{p}_\theta)$ and $H(p)$ is constant in θ ,

$$\arg \min_{\theta} H(p, \hat{p}_\theta) = \arg \min_{\theta} D_{\text{KL}}(p \| \hat{p}_\theta).$$

So minimizing the “cross-entropy loss” is *identical* to minimizing $D_{\text{KL}}(p \| \hat{p}_\theta)$ — the cross-entropy differs from the KL divergence only by the additive constant $H(p)$, which plays no role in optimization. This is why the two terms are used almost interchangeably in ML, even though they are only equal (not merely equivalent for optimization) in the special case $H(p) = 0$, i.e. hard one-hot labels; for soft labels ($H(p) > 0$, e.g. label smoothing or knowledge distillation) cross-entropy and KL differ by this nonzero constant, though both still have the same minimizer in θ .